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Prototype Build Plan — With Directions

A practical step-by-step guide for assembling, wiring, programming, calibrating and testing the two smart gloves.

Important: This guide uses the exact component roles described in your project. Exact ESP32 GPIO numbers, TFT pins and MAX98357A pins depend on the specific modules/boards you have. Before making permanent connections, verify the pin labels printed on your actual boards and their datasheets.

System flow

Left glove: 5 flex sensors + MPU6050 → ESP32 → TFT display

Right glove: 5 flex sensors + MPU6050 → ESP32 → MAX98357A → 3 W speaker

Mobile app: speech → text → wireless communication → left-glove TFT

1. WORKSPACE SETUP

Step 1 — Make two work areas

Direction: Put the left-glove components on one side and the right-glove components on the other. Keep batteries separate until power testing.

Step 2 — Label everything

Direction: Label the five flex sensors T, I, M, R, L for thumb, index, middle, ring and little finger. Label the two ESP32 boards LEFT and RIGHT.

Step 3 — Prepare a wiring record

Direction: Draw a table with component, ESP32 pin, power, ground and purpose. Do not rely on memory when many wires are connected.

Step 4 — Keep power disconnected

Direction: During signal wiring, keep batteries and USB power disconnected unless you are intentionally testing the circuit.

2. FLEX SENSOR WIRING AND DIRECTION

Step 5 — Understand the flex sensor

Direction: A flex sensor changes resistance when it bends. The ESP32 cannot directly measure resistance, so use a voltage divider.

Step 6 — Build one test channel first

Direction: Connect 3.3 V to one end of the flex sensor. Connect its other end to the ADC measurement point. Connect a fixed resistor from the ADC point to GND.

Step 7 — Connect the ADC point

Direction: Connect the voltage-divider midpoint to an ADC-capable ESP32 input. Keep the sensor signal wire short and secure.

Step 8 — Repeat for five fingers

Direction: Build five separate voltage-divider channels. Never connect multiple sensor outputs together.

Step 9 — Mount the sensors

Direction: Place each sensor along the finger so it bends naturally with the finger. Keep the sensor's sensitive section away from sharp folds.

Step 10 — Check orientation

Direction: Before fixing the sensor, bend the finger slowly. Make sure the sensor bends in the intended direction and the cable does not pull on the sensor.

Step 11 — Test one sensor

Direction: Power the ESP32 and print the ADC reading. Record the value with the finger straight, half bent and fully bent.

Step 12 — Calibrate all five

Direction: Repeat the measurement for thumb, index, middle, ring and little finger. Take several readings for each position.

Step 13 — Create thresholds

Direction: Use the observed ranges to define straight, bent and intermediate states. Use ranges rather than a single exact number.

3. MPU6050 WIRING AND DIRECTION

Step 14 — Identify the MPU6050 pins

Direction: Find VCC, GND, SDA and SCL on your actual module. Follow the module's voltage requirements.

Step 15 — Connect power

Direction: Connect the module's required supply and GND. For a typical ESP32 setup, the sensor logic is commonly operated at 3.3 V, but verify your module.

Step 16 — Connect I²C

Direction: Connect SDA to the ESP32 SDA pin and SCL to the ESP32 SCL pin selected in your firmware.

Step 17 — Run an I²C scan

Direction: Upload an I²C scanner and verify that the MPU6050 appears at its expected address. If it does not, check power, ground and SDA/SCL wiring.

Step 18 — Test acceleration

Direction: Print AX, AY and AZ. Keep the glove still, then tilt it forward, backward and sideways.

Step 19 — Test rotation

Direction: Print GX, GY and GZ. Rotate the glove around different axes and confirm that the readings respond.

Step 20 — Calibrate

Direction: Keep the glove motionless and calculate or record the offsets required by your software.

Step 21 — Repeat for the second glove

Direction: Perform the same wiring, address check and calibration on the right glove.

4. LEFT DISPLAY GLOVE

Step 22 — Plan physical placement

Direction: Place the ESP32 where it will not interfere with the wrist. Place the TFT on the back of the hand where it is visible to the wearer.

Step 23 — Wire the TFT

Direction: Identify the TFT interface and connect power, ground and its communication/control pins according to the exact module pinout.

Step 24 — Test the display alone

Direction: Upload a simple program that displays HASTH VANI, READY and a test message.

Step 25 — Add flex and MPU6050 inputs

Direction: Run the sensor-reading program and display the live sensor values or a short diagnostic screen.

Step 26 — Create a gesture record

Direction: Combine five flex readings and the relevant MPU6050 orientation values into one data structure.

Step 27 — Recognize one gesture

Direction: Start with HELLO. Record several examples and identify the sensor ranges that distinguish it.

Step 28 — Add the message

Direction: When HELLO is recognized, display HELLO on the TFT.

Step 29 — Add more gestures

Direction: Add WATER, HELP, THANK YOU, YES, NO and other predefined messages chosen by your team.

Step 30 — Add an unknown state

Direction: If no gesture matches reliably, display UNKNOWN or READY rather than producing a random message.

5. RIGHT SPEAKER GLOVE

Step 31 — Mount the second ESP32

Direction: Place it securely and leave enough room for the USB connector and wiring.

Step 32 — Mount the speaker

Direction: Position the speaker so its sound outlet is not covered by the glove or mounting material.

Step 33 — Connect MAX98357A

Direction: Identify VIN/VCC, GND, BCLK, LRC/WS and DIN on the exact amplifier board. Connect the ESP32 I²S signals according to your firmware and module.

Step 34 — Connect speaker

Direction: Connect the speaker to the amplifier's speaker output using the module's marked terminals. Check polarity and avoid shorting the outputs.

Step 35 — Test audio

Direction: Before adding gesture recognition, play a known audio test. Verify volume, clarity and temperature.

Step 36 — Add the second glove sensors

Direction: Connect its five flex sensors and MPU6050 using the same approach as the display glove.

Step 37 — Recognize one gesture

Direction: Test HELLO first. Confirm that the second ESP32 identifies the intended gesture.

Step 38 — Trigger audio

Direction: Map the recognized gesture to the corresponding audio message and play it through the MAX98357A and speaker.

6. GESTURE RECOGNITION DIRECTIONS

Step 39 — Collect real samples

Direction: Wear the glove and perform each intended gesture repeatedly. Save the sensor values for successful and unsuccessful attempts.

Step 40 — Normalize sensor behavior

Direction: If needed, map each flex reading to a normalized range using its calibrated straight and bent limits.

Step 41 — Use multiple samples

Direction: Do not accept a gesture from one noisy reading. Check several consecutive readings and accept the gesture only when they remain consistent.

Step 42 — Combine hand shape and orientation

Direction: Use both flex-sensor states and MPU6050 orientation where necessary so similar finger patterns can be separated.

Step 43 — Add a cooldown

Direction: After a gesture is accepted, briefly prevent immediate repeated triggering. This reduces duplicate messages.

Step 44 — Test transitions

Direction: Move from one gesture to another and check whether the system incorrectly recognizes intermediate positions.

7. MOBILE APP AND TWO-WAY COMMUNICATION

Step 45 — Connect the app

Direction: Use the wireless method already selected for your project, such as Bluetooth or Wi-Fi.

Step 46 — Show connection status

Direction: Display connected/disconnected state so the user knows whether communication is available.

Step 47 — Send recognized messages

Direction: Send the recognized gesture/message to the app for history or display.

Step 48 — Implement speech-to-text

Direction: Capture speech from the communication partner's phone microphone and convert it into text using the app's speech-recognition function.

Step 49 — Send text to the left glove

Direction: Transmit the resulting text to the display glove.

Step 50 — Display received text

Direction: Show the received message clearly on the TFT.

Step 51 — Test the complete reverse direction

Direction: Speak a short sentence, verify the app's transcription, verify wireless transmission and confirm that the TFT displays the result.

8. POWER AND SAFE PHYSICAL ASSEMBLY

Step 52 — Test power modules separately

Direction: Check the TP4056 charging module and battery arrangement before connecting the complete glove.

Step 53 — Insulate exposed joints

Direction: Cover solder joints and exposed conductors with suitable insulation. Prevent accidental shorts.

Step 54 — Add strain relief

Direction: Secure wires near flex sensors, the wrist and board connectors so movement does not pull solder joints.

Step 55 — Route wires

Direction: Keep wires along the glove surface and away from moving finger joints as much as possible.

Step 56 — Secure the boards

Direction: Use a secure but serviceable mounting method. Do not permanently seal the electronics until final testing is complete.

Step 57 — Check comfort

Direction: Wear the glove and slowly move every finger. Confirm that no board, wire or battery causes uncomfortable pressure.

Step 58 — Check heat

Direction: During a full-power test, touch-check only where safe and monitor components for abnormal heating. Disconnect power if anything becomes unexpectedly hot.

9. COMPLETE TESTING

Step 59 — Test left glove alone

Direction: Run all five flex sensors, MPU6050 and TFT together.

Step 60 — Test right glove alone

Direction: Run all five flex sensors, MPU6050, amplifier and speaker together.

Step 61 — Test gesture accuracy

Direction: Perform every supported gesture multiple times and record correct and incorrect recognitions.

Step 62 — Test false triggers

Direction: Move naturally without intentionally making a gesture. Check whether the system produces unwanted messages.

Step 63 — Test response time

Direction: Measure the approximate time from completing a gesture to seeing or hearing the output.

Step 64 — Test wireless range

Direction: Walk the distance required for your demonstration and verify that communication remains stable.

Step 65 — Test restart behavior

Direction: Turn the system off and on again. Confirm that both gloves return to a usable state.

Step 66 — Test battery operation

Direction: Run the prototype from its intended battery arrangement and observe operating stability.

Step 67 — Test the two-way demo

Direction: Gesture → recognition → voice/text, then spoken response → speech-to-text → wireless message → TFT.

10. FINAL ASSEMBLY

Step 68 — Freeze the wiring

Direction: After all tests pass, photograph the wiring and document every connection before final dressing.

Step 69 — Secure components

Direction: Fix the ESP32, sensors, display, amplifier and speaker in their final positions.

Step 70 — Label important wires

Direction: Label power, ground, I²C and audio lines where practical to simplify future repairs.

Step 71 — Protect connectors

Direction: Secure loose connectors so normal glove movement does not disconnect them.

Step 72 — Final calibration

Direction: Perform one final calibration while wearing the finished glove, because mounting position can change sensor readings.

Step 73 — Save firmware

Direction: Keep a copy of the final ESP32 code and the last known-good version.

Step 74 — Record limitations

Direction: Document the supported gestures, approximate recognition behavior, battery limitations and known failure cases.

11. 10 AM–6 PM BUILD-DAY DIRECTIONS

10:00–10:20

Direction: Set up two workstations, label components, photograph starting condition and assign team roles.

10:20–11:10

Direction: Mount and wire the ten flex sensors. Test each sensor before permanent fixing.

11:10–12:00

Direction: Calibrate all flex sensors and record threshold values.

12:00–12:40

Direction: Wire and test both MPU6050 modules.

12:40–1:10

Direction: Save calibration data and take a short break.

1:10–2:00

Direction: Implement and test the first gesture-recognition rules.

2:00–3:00

Direction: Integrate the left glove with TFT and test displayed messages.

3:00–4:00

Direction: Integrate the right glove with MAX98357A and speaker.

4:00–4:40

Direction: Connect the mobile application and implement/test speech-to-text communication.

4:40–5:20

Direction: Run the complete two-way system and fix the most important failures.

5:20–5:50

Direction: Secure wiring, mount components and perform final calibration.

5:50–6:00

Direction: Run the demonstration once, save code/data, photograph the final prototype and pack all parts.

TEAM ROLE PLAN

Member	Primary responsibility
1	Left-glove flex sensors, calibration and physical mounting
2	Right-glove flex sensors, calibration and physical mounting
3	ESP32 firmware and gesture-recognition logic
4	TFT/display glove and wireless message display
5	MAX98357A, speaker and right-glove audio system
6	Mobile app, speech-to-text, testing and documentation

FINAL CHECKLIST

- All 10 flex sensors tested and calibrated.
- Both MPU6050 modules detected and calibrated.
- Left ESP32 + TFT working.
- Right ESP32 + MAX98357A + speaker working.
- Gesture recognition tested for every supported gesture.
- Unknown/no-gesture state works.
- Mobile speech-to-text works.
- Text reaches the display glove.
- Two-way communication demonstrated.
- Battery and charging system checked safely.
- Wires secured and exposed connections insulated.
- Final firmware backed up.
- Supported gestures and limitations documented.

TROUBLESHOOTING QUICK GUIDE

Problem	First checks
Flex values do not change	Check voltage divider, ADC pin, ground, sensor orientation and wire continuity.
MPU6050 not detected	Check VCC/GND, SDA/SCL, I ² C address and pull-ups/module configuration.
TFT blank	Check power, ground, interface type, CS/DC/RST pins and display initialization.
Speaker silent	Check I ² S pins, MAX98357A power, speaker connection and audio sample configuration.
Random gestures	Recalibrate flex sensors, widen/narrow thresholds appropriately, use multiple samples and add a cooldown.
Wireless connection drops	Check power stability, distance, reconnection logic and whether both devices use the same communication settings.